IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

MSc and EEE PART III/IV: MEng, BEng.and ACGI

MICROWAVE TECHNOLOGY

Wednesday, 13 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer question ONE.

Answer TWO out of the remaining THREE questions.

Question ONE carries a weighting of 50%.
The rest carry a weighting 25% each

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Lucyszyn
Second Marker(s) : K. Fobelets

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Special instructions for invigilators

This is an OPEN BOOK examination.

Special instructions for students

There are 4 (four) questions in total.

Question 1 is COMPULSORY and carries a weighting of 50%.

Answer 2 (two) out of the remaining 3(three) questions, each carrying a weighting of 25%.

All variables and abbreviations have their usual meaning.

1.

a) Two 500 μm thick, microstrip-based integrated circuits are conductively bonded onto a common ground plane. Consider a 1 mm long horizontal, 2 mil diameter gold bond wire (with DC bulk conductivity of 4.517 x 10^7 S/m), connecting the RF bond pads on both chips. The ideal cross-section of this bond-wire chip-to-chip interconnect scenario is illustrated in Fig. 1.

Draw a simplified lumped-element equivalent circuit model for this chip-to-chip interconnect. Very briefly, describe this circuit in terms of its frequency response.

[4]

b) Draw a simplified distributed-element equivalent circuit model for the bond-wire part of this chip-to-chip interconnect. With an operation at 2 GHz, justify the modelling of a 1 mm long bond wire with the use of a distributed-element equivalent circuit model.

[4]

c) Unless otherwise stated, calculate the following at 2 GHz and include the units; clearly stating any simplifying assumptions:

i) Total DC resistance. [4]

ii) Distributed resistance. [4]

iii) Distributed inductance (L_{pul} in Figure 1). [3]

iv) Distributed conductance. [2]

v) Distributed capacitance (C_{pul} in Figure 1). [3]

vi) Lossless characteristic impedance. [3]

vii) Lossy characteristic impedance. [3]

viii) Lossless phase velocity. [2]

ix) Lossless phase constant. [2]

x) Attenuation constant due to the conductors only. [2]

xi) Attenuation constant due to the air dielectric only. [2]

xii) Total attenuation constant and insertion loss in dB/mm. [3]

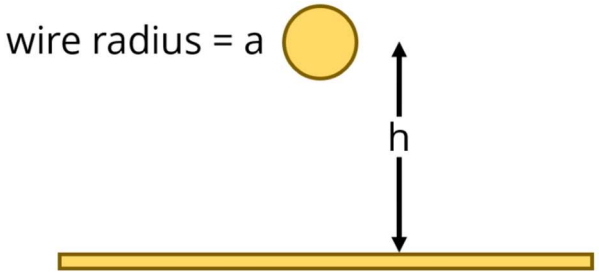
xiii) Q-factor associated with the conductors only. [2]

xiv) Q-factor associated with the air dielectric only. [2]

xv) Overall Q-factor. [2]

d) Sketch a realistic implementation of a bond-wire chip-to-chip interconnect and briefly explain why the models in 1(a), 1(b) and 1(c) are not realistic at millimetre-wave frequencies.

[3]



$$L_{pul} = \frac{\mu_0 \mu_r}{2\pi} \cosh^{-1}\left(\frac{h}{a}\right)$$

$$C_{pul} = \frac{2\pi\varepsilon_0 \varepsilon_r}{\cosh^{-1}\left(\frac{h}{a}\right)}$$

Figure 1. Simplified cross-sectional illustration of a bond-wire chip-to-chip interconnect.

2.

Consider a metallic wire mesh sheet.

i) Define a metamaterial. What are the conditions needed for the wire mesh to be a metamaterial?

[5]

ii) Give an alternative name for this sheet.

[1]

iii) Sketch the transmittance frequency response and state how this compares to a filter.

[5]

iv) How is shielding effectiveness defined, in terms of power?

[2]

v) If shielding effectiveness is equal to $(2l/\lambda)^2$, where l is the maximum geometric dimension of the hole, calculate both the percentage transmittance and dB screening effectiveness values for a wire mesh having square apertures with a side length of 2.54 mm in air at 3 GHz. Does the maximum aperture length meet the dimensional condition for a metamaterial?

[5]

vi) Using your answer in 2(v), and assuming that the wire is a PEC and ignoring diffraction effects, calculate the associated percentage reflectance and return loss values of the mesh at 3 GHz.

[3]

vii) Give two examples of where a wire mesh is used at microwave frequencies and briefly explain their effects of frequency filtering.

[4]

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3.

At 2.45 GHz, human skin has a dielectric constant of 44 and a conductivity of 1.85 S/m. Calculate the following for human skin at 2.45 GHz, using the appropriate expressions and showing units:

i) Wavelength. [2]

ii) Imaginary part of the relative permittivity. [3]

iii) Loss tangent. [2]

iv) Quality factor. [2]

v) Attenuation constant. [4]

vi) Phase constant. [4]

vii) Power attenuation in dB per wavelength. [4]

viii) Intrinsic impedance. [4]

4.

A simple low frequency microwave FET amplifier can be realised by employing conventional microstrip transmission lines for its interconnects, impedance matching and biasing networks.

a) Describe the main problem with conventional microstrip circuits when trying to ground the source connection of a FET. Give two examples of catastrophic circuit failure. Also, state how a conventional microstrip circuit can be modified to reduce this problem and why.

[5]

b) Compare and contrast the manufacture of TFMS with conventional microstrip. Use suitable illustrations, showing variables for the main cross-sectional dimensions.

[4]

c) Write simple equations, to a first-order of approximation, to justify each of the following claims:

i) the characteristic impedances of a TFMS and conventional microstrip line can be equal.

[5]

ii) losses in a TFMS are higher than in a corresponding conventional microstrip line.

[5]

d) Figure 2 shows the photograph of a MMIC amplifier, employing TFMS transmission lines. By inspection, draw the equivalent circuit model of this amplifier and state the role of the two longest TFMS transmission lines. What kind of amplifier topology is this?

[6]

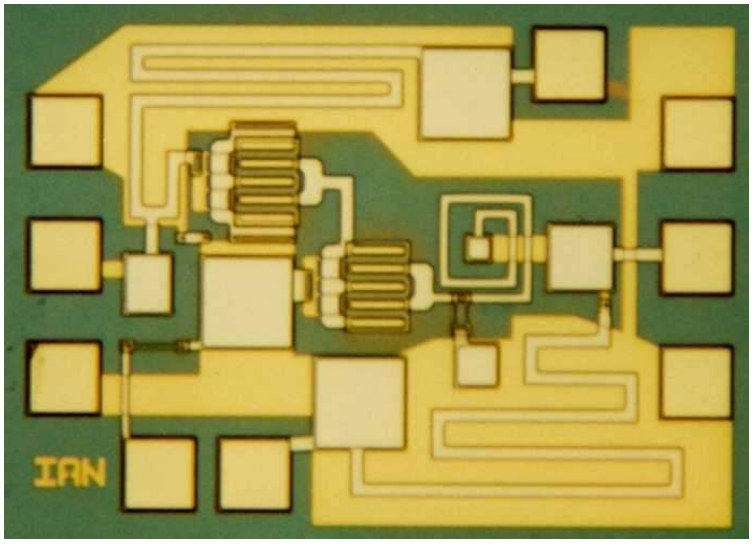


Figure 2. MMIC amplifier.